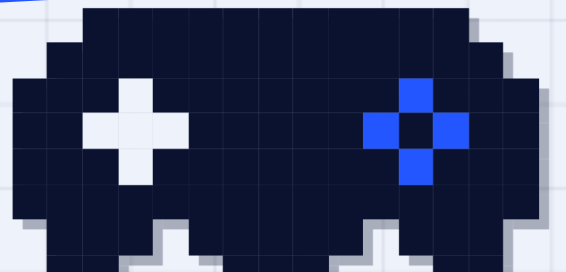
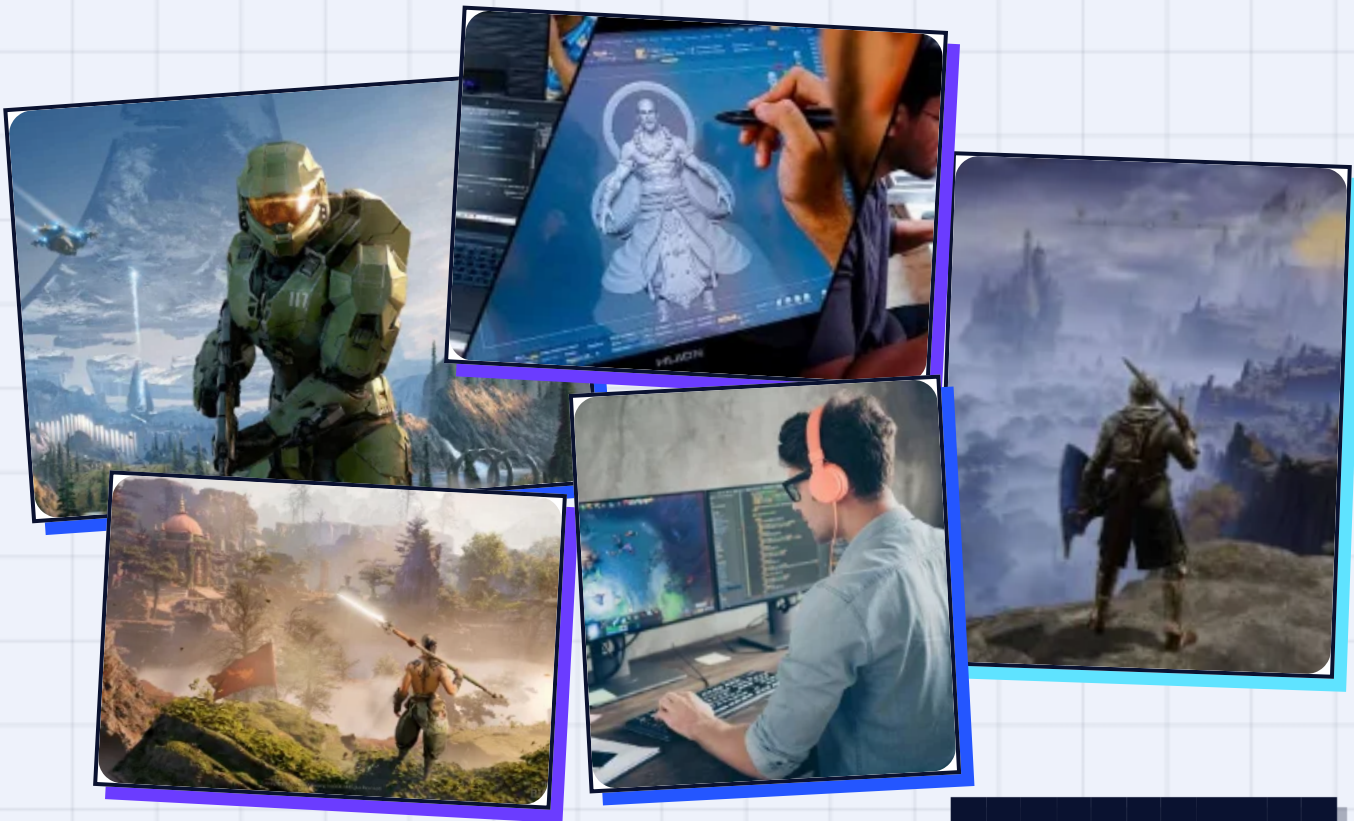


PROFESSIONAL PROGRAM / 2026

GAME DEVELOPMENT

Build Games. Learn the Technology. Create Your Portfolio.



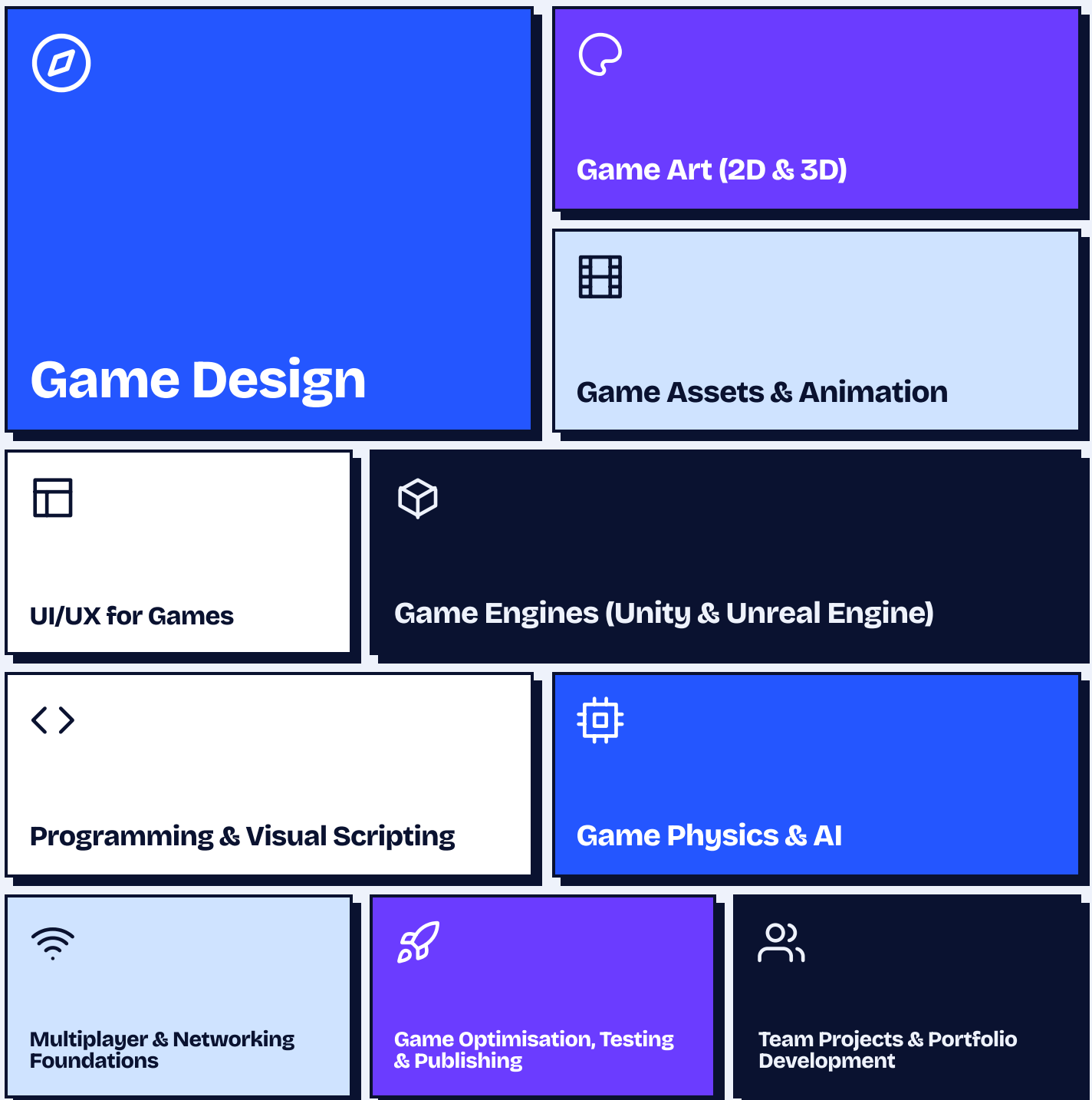
FROM FIRST IDEA TO **FINISHED** GAME.

Learn how games are conceived, designed, programmed, illustrated, animated, tested and published, from small exercises to a capstone game project.

**Conceived****Programmed****Animated****Published****Designed****Illustrated****Tested**

The program moves students through the full game development journey: from idea and artwork, through 2D and 3D assets and animation, into game engines, programming and game systems, and finally into optimisation, testing and a complete, presentable project.

ONE PROGRAM. TEN DISCIPLINES.



7 SEMESTERS. ONE FINISHED PORTFOLIO.

Semester 01 — Working with Artworks

10 TOPICS

01 Working on the idea for the game

02 Assembling the idea into a game concept

03 Assembling different artwork references

04 Understanding the artwork style

05 Designing rough assets for the game

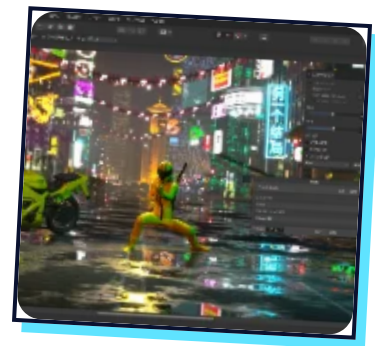
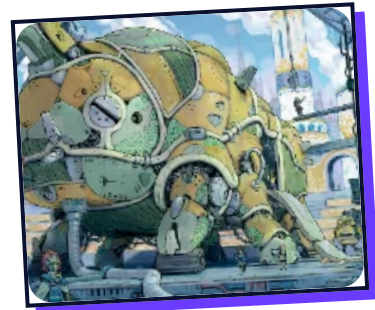
06 Understanding the importance of storyboarding

07 2D content development

08 Designing characters for the game

09 Environment & background study for games

10 Designing rough levels



Skill development: Students should understand the importance of artwork for game development, animation, VFX and the broader graphics industry. The module establishes the foundation for understanding how a game idea becomes visual content and a project concept.



- 01 Understanding colour theory
- 02 2D asset creation pipeline
- 03 Creating assets in 2D software
- 04 Working with Adobe Photoshop
- 05 Understanding project flow in Photoshop
- 06 Understanding graphic art style
- 07 Working on 2D game background/UI assignments
- 08 Working with Adobe Illustrator
- 09 Designing complete 2D game art with Photoshop, Illustrator & Animate
- 10 Learning UI/UX concepts in detail
- 11 Group live project

Skill development: Students should develop strong graphic visualisation skills and understand how to create complete graphical assets and interfaces for game projects, with a focus on practical visual production rather than purely theoretical design.

SEMESTER

03

8 TOPICS

Mastering 2D Game Assets with Animation



- 01 Learning to match and compose 2D art & graphics in a scene

- 02 Working with industry-standard applications

- 03 Working with Adobe Animate/Spriter for game animations

- 04 Learning to render game assets & animation for a game engine

- 05 Understanding character sprite sheets for games & animation

- 06 Working with sprite sheets & creating animations for games

- 07 Scene and cinematic compositing

- 08 Working on 2D games with Unity & Unreal Engine

Skill development: Students should learn how game assets are used for games and animations and how those assets move through the development process into actual game projects, including live/project-based application.

SEMESTER

04

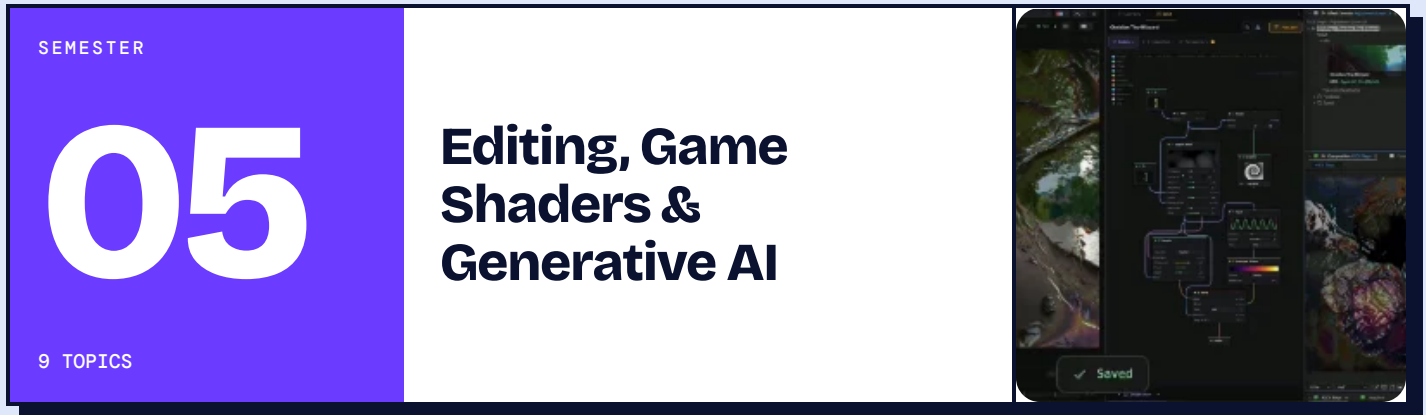
11 TOPICS

Mastering 3D Game Assets with Animatics



- 01 Learning to create, match & compose 3D assets & art
- 02 Working with industry-standard applications
- 03 Working with Maya/Blender & Mudbox/ZBrush
- 04 Creating assets for 3D games
- 05 Working on game textures & game background art
- 06 Creating cinematics for 3D games
- 07 Creating animation for 3D games
- 08 Learning to make prefabs
- 09 Assembling assets as a group
- 10 Scene and cinematic compositing
- 11 Working on 3D games with Unity & Unreal Engine

Skill development: Students should learn how to apply 2D and 3D skills to games and multimedia projects and create 3D assets for game development.



- 01 Post-production techniques for games
- 02 Learning to edit & compose scenes
- 03 Learning After Effects and DaVinci Resolve
- 04 PFTrack for camera tracking, rotoscopy & 3D tracking
- 05 Applying effects in game engines like Unity and Unreal Engine
- 06 Understanding generative AI for asset creation with AI chatbots
- 07 Working with shaders & particles
- 08 Learning to create complex shaders & particles for games
- 09 Live group project

Skill development: VFX skills should be developed as an important supporting skill for games and multimedia projects. The module connects post-production, effects, shaders, particles and modern AI-assisted asset workflows.



01 Understanding visual scripting & game programming languages

02 Learning Unity Visual Scripting with C#

03 Learning Unreal Blueprint with C++

04 Coding advanced C++ & C# logic for high-end games

05 Learning to integrate 2D & 3D games using C# in Unity

06 Learning to integrate 2D & 3D games using C++ in Unreal Engine

07 Understanding game logic, physics, AI movement & networking

08 Understanding the pipeline to link cinematics & games

Skill development: The programming module covers the programming and scripting side of game development. Students develop the ability to work with C#, C++, Unity, Unreal Engine, visual scripting, game logic, physics, AI and networking concepts.

SEMESTER

07

9 TOPICS

Creating Scalable Games & Apps



- 01 Assembling created assets in the game engine
- 02 Learning to use GitHub and work in a team
- 03 Learning cross-platform basics
- 04 Developing playable games across genres such as platform, puzzle, shooter, racing, match-3, AI-based, social and others
- 05 Applying AI to enemy behaviour
- 06 Game level development & linking
- 07 Learning the basics of networking and a scalable project pipeline
- 08 Game performance optimisation
- 09 Game testing and debugging

Skill development: Students should develop the skills required to create playable games across different genres and understand the practical processes involved in problem solving, applying AI, networking concepts, optimisation, testing, debugging and collaborative development.

LEARN IT. THEN BUILD IT.

**01**

Understand

Learn the concepts behind game design, art, engines and programming before building with them.

02

Practise

Work through guided exercises in art, animation, engines and code to turn concepts into skills.

03

Build

Apply those skills across individual assignments, group projects and full game builds.

04

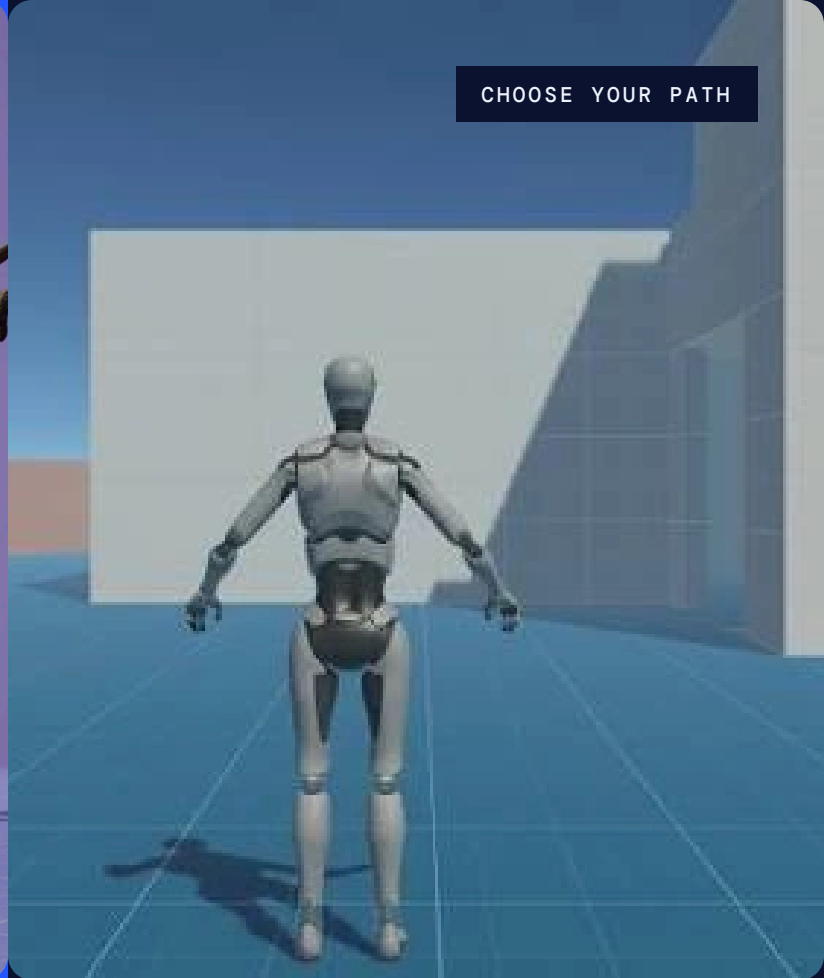
Show

Bring assets, animation, engine work and code together into playable, presentable projects.

05

Prepare

Build the practical confidence and portfolio needed for the next step, whether that's further learning or a career.



UNREAL PATH

UNREAL

Go deeper on cinematic-quality visuals and high-performance C++ systems.

Unreal Engine

Blueprint

C++

UNITY PATH

UNITY

Go deeper on fast iteration, cross-platform builds and approachable C# scripting.

Unity Engine

Visual Scripting

C#

FIVE PROJECTS. EACH ONE BIGGER.

PROJECT 01

Game Concept & Artwork

Game idea

Concept

Artwork direction

Rough assets

Characters

Environments

Rough levels

PROJECT 02

2D Game Project

2D assets

UI

Backgrounds

Animation

Game scenes

Unity / Unreal

PROJECT 03

3D Game Project

3D assets

Textures

Environments

Animation

Cinematic elements

Prefabs

Unity / Unreal

PROJECT 04

Gameplay & Programming Project

Game logic

Programming

Visual scripting

Physics

AI

Gameplay systems

PROJECT 05

Advanced Game Project

Scalable game structure

GitHub / team workflow

Cross-platform concepts

Optimisation

Testing

Debugging

Networking foundations

YOUR TOOLKIT.

Game Engines

Unity

Unreal Engine

Programming

C#

C++

Visual Scripting

Unity Visual Scripting

Unreal Blueprint

3D & Art

Maya

Blender

Mudbox

ZBrush

2D & Creative

Adobe Photoshop

Adobe Illustrator

Adobe Animate

Post-Production & VFX

After Effects

DaVinci Resolve

PFTrack

Collaboration

GitHub

AI

Generative AI

AI-assisted asset creation

AI-assisted game workflows

WHERE A GAME PORTFOLIO CAN POINT.

Potential directions a portfolio from this program can support. Not a placement or job guarantee.

/ **Game Designer**

/ **Game Developer**

/ **Unity Developer**

/ **Unreal Developer**

/ **Gameplay Programmer**

/ **Game Artist**

/ **2D Game Artist**

/ **3D Game Artist**

/ **Game UI/UX Designer**

/ **Technical Artist**

/ **Game Animator**

/ **VFX Artist for Games**

/ **Level Designer**

/ **Junior Game Programmer**

/ **Game QA / Testing**

Portfolio

By the end of the program, a student can put together a portfolio spanning game concepts and artwork, 2D and 3D assets and animation, engine-built playable levels, programmed gameplay systems and a larger, team-built project: real, presentable evidence of what they can build, not just what they have studied.

GAME DEVELOPMENT PROFESSIONAL PROGRAM

30

MONTHS OF LEARNING

7

SEMESTERS

66

TOPICS

1

CAPSTONE GAME PROJECT

- 30 months of learning
- A capstone game project
- Exposure to industry-standard game development tools



ENQUIRE NOW →

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Professional Game Development & Design. Learn how games are conceived, designed, programmed, illustrated, animated, tested and published, from small exercises to a capstone game project.

ENQUIRE ABOUT THE PROGRAM →

VIEW PROGRAM ONLINE

TECHNO GURUKUL

Learning Institute

Office No. 305, Platinum Plaza, opp. Ramayan Bungalow, next to
Jain Oswal Boarding, Tilak Wadi, Police Staff Colony, Nashik,
Maharashtra 422002

CONTACT

admission@technogurukul.com